

QCE Chemistry Units 3&4

Paper 2

Student's Name: _____

Teacher's Name: _____

Time allowed

- Perusal time – 10 minutes
- Working time – 90 minutes

General instructions

- Answer all questions in this question and response booklet.
- Write using black or blue pen.
- QCAA-approved calculator permitted.
- Formula and data booklet provided.
- Planning paper will not be marked.

Section 1 (65 marks)

- 8 short response questions

Students are advised that this is a trial examination only and cannot in any way guarantee the content or the format of the 2022 QCE Chemistry Units 3&4 Written Examination.

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SECTION 1

Instructions

- If you need more space for a response, use the additional pages at the back of this booklet.
 - On the additional pages, write the question number you are responding to.
 - Cancel any incorrect response by ruling a single diagonal line through your work.
 - Write the page number of your alternative/additional response, i.e. See page ...
 - If you do not do this, your original response will be marked.
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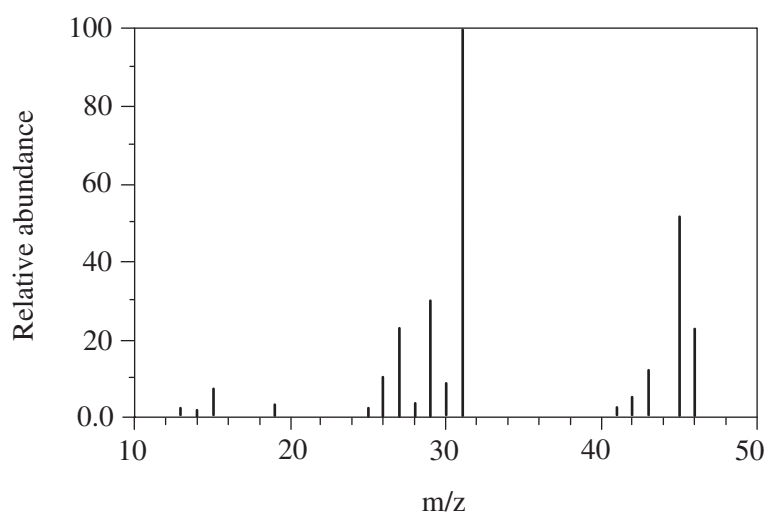
DO NOT WRITE ON THIS PAGE

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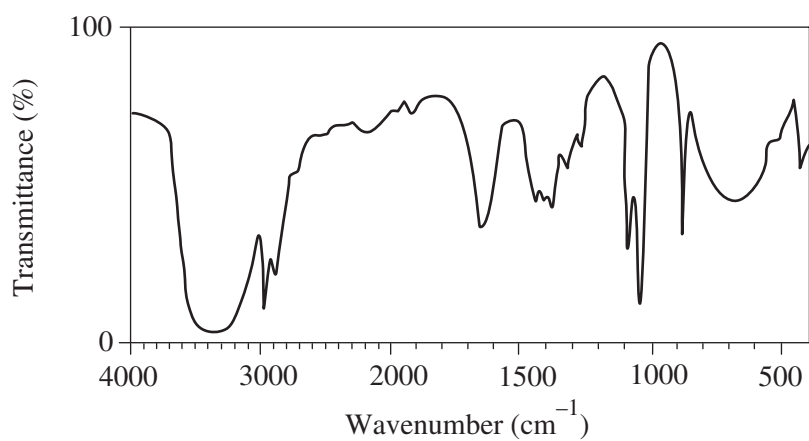
QUESTION 1 (8 marks)

The mass and infrared spectra for an alcohol are shown.

Mass spectrum



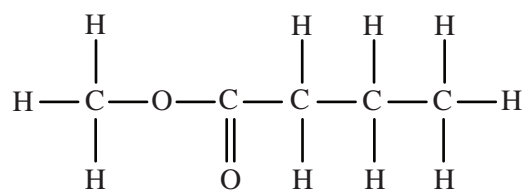
Infrared spectrum



- a) Analyse both spectra to deduce the structural formula of the alcohol. Explain your reasoning.

[6 marks]

- b) The diagram shows an ester that was synthesised.



Determine whether the alcohol deduced in 1a) was or was not used to synthesise the ester. Explain your reasoning.

[2 marks]

- b) Calculate the overall emf of the cells. Show your working.

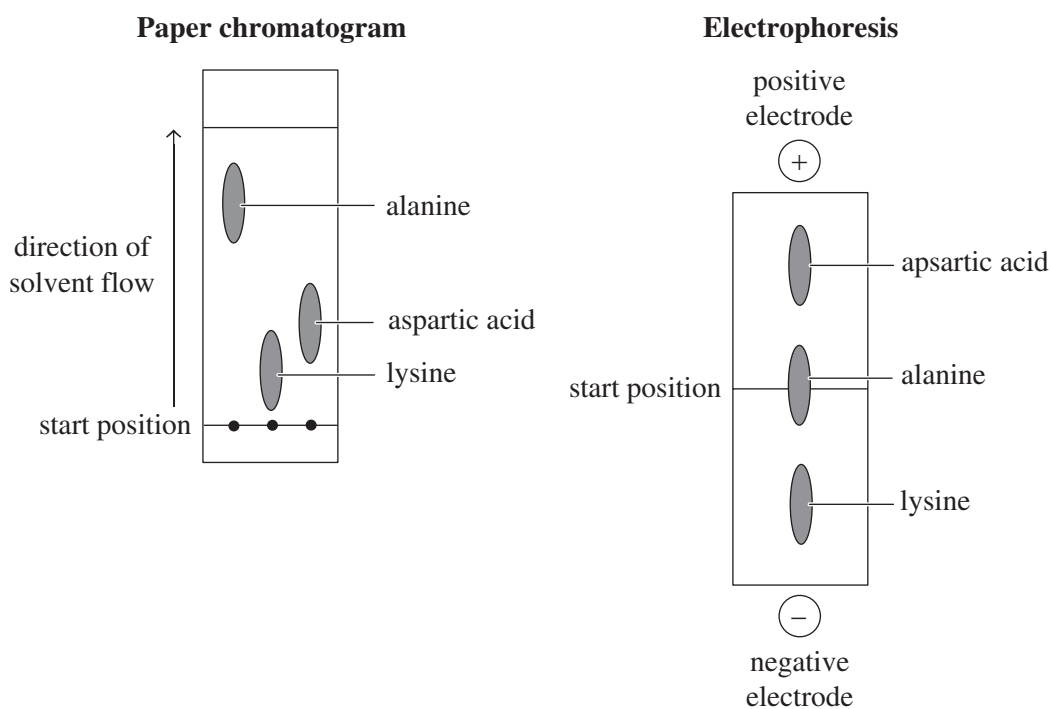
[2 marks]

emf = _____ V

QUESTION 3 (8 marks)

A scientist wished to separate the amino acids alanine, aspartic acid and lysine.

The diagrams show a paper chromatogram of a mixture of the amino acids and the result of an electrophoresis experiment using the same mixture in a buffer with a pH of 6.1.



- a) In the paper chromatogram, alanine is separated from the other amino acids by a large distance and there is poor separation between lysine and aspartic acid.

Explain why this occurs.

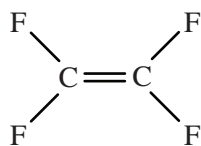
[3 marks]

- b) Explain why there is good separation between lysine and aspartic acid in the electrophoresis experiment. Include diagrams in your response. *[3 marks]*

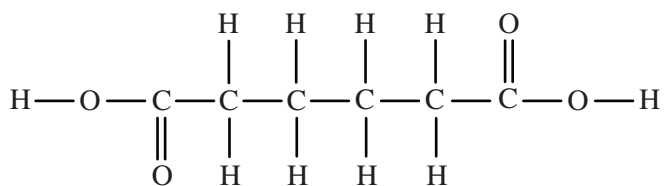
- c) Explain why alanine hardly moved in the electrophoresis experiment. Include a diagram in your response. *[2 marks]*

QUESTION 4 (9 marks)

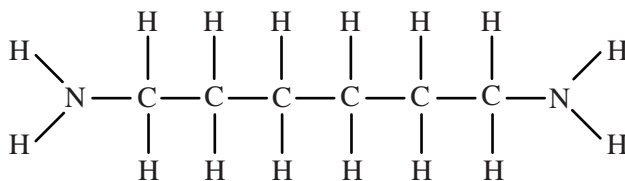
Polymers are usually formed by joining smaller units (monomers) together. Three monomers are shown.



tetrafluoroethylene



adipic acid



hexamethylene diamine

- a) Draw and name the polymer formed by joining many tetrafluoroethylene units together. [2 marks]

IUPAC name _____

When adipic acid and hexamethylene diamine are joined, they form the polymer nylon 6,6.

- b) Draw the polymer formed by joining many adipic acid and hexamethylene diamine units together.

[1 mark]

- c) Complete the table below to compare the polymers from 4a) and 4b).

[4 marks]

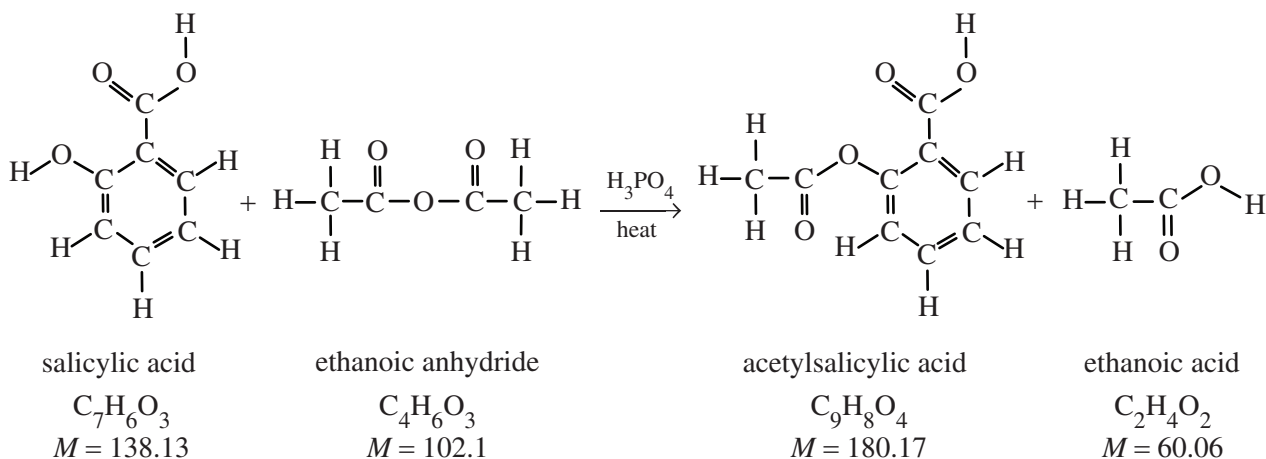
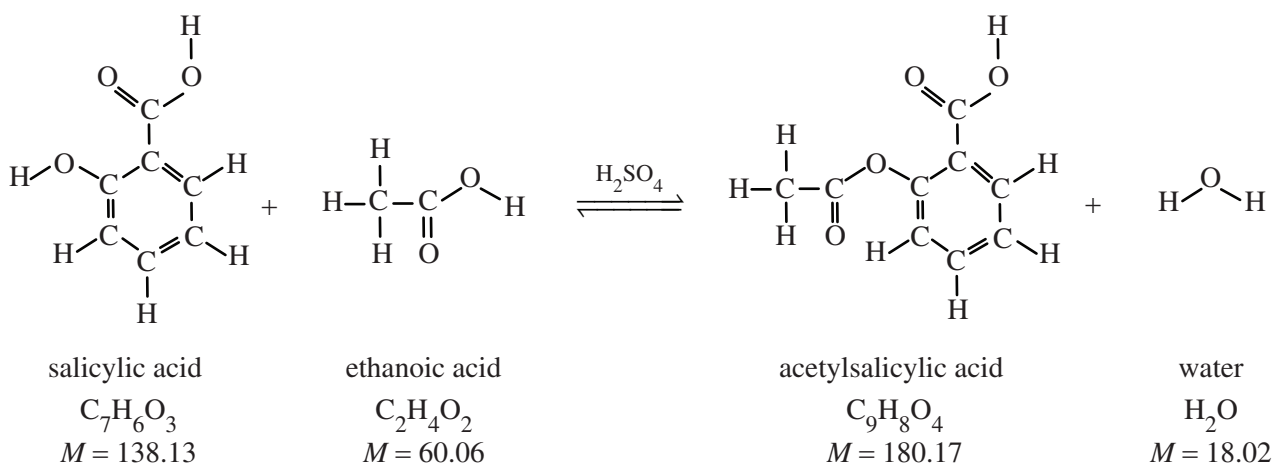
Monomer(s)	Type of reaction to form polymer	Does the polymer formed have alternating units?
tetrafluoroethylene		
adipic acid and hexamethylene diamine		

- d) i) What does '6,6' in the name nylon 6,6 indicate? *[1 mark]*

- ii) Propose why nylon 6,6 forms strong fibres. Refer to intermolecular forces in your response. *[1 mark]*

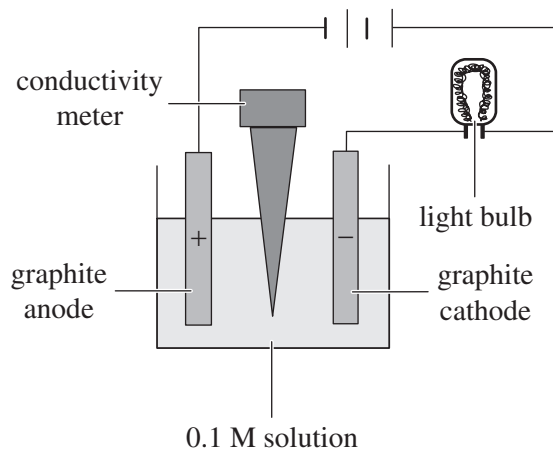
QUESTION 5 (7 marks)

Aspirin (acetylsalicylic acid, $C_9H_8O_4$) can be prepared through two different reactions. The reactions are shown.

Reaction 1**Reaction 2**

QUESTION 6 (11 marks)

The diagram shows an experiment set-up that was used to investigate the relative strengths of acids and bases in aqueous solution. The power supply was connected to two graphite electrodes.



The brightness of the light bulb and relative electrical conductance for each solution are recorded in the table.

Solution (0.1 M)	Light bulb brightness	Relative electrical conductance	pK_a	pK_b
ethanoic acid (CH_3COOH)	dim	4.2	4.74	–
hydrochloric acid (HCl)	very bright	11.7	–5.90	–
nitric acid (HNO_3)	bright	6.8	–1.37	–
sulfuric acid (H_2SO_4)	bright	8.8	–2.00	–
hydroiodic acid (HI)			–9.30	–
sodium hydroxide (NaOH)	very bright	10.6	–	–0.56
methylamine (CH_3NH_2)		3.1	–	3.34
ammonia (NH_3)		4.2	–	4.75

- a) Complete the table above by indicating the brightness of the bulbs and estimating the relative electrical conductance of HI. [4 marks]
- b) i) Identify which acid has the lowest pH. Explain your reasoning. [2 marks]

- ii) Determine the strongest base in terms of degree of ionisation at equilibrium in aqueous solution. Explain your reasoning.

[2 marks]

- c) Describe the relationship between the concentration of hydrogen ions (H^+), electrical conductivity and light bulb brightness.

[1 mark]

- d) An unknown solution, which was thought to be an acid, was tested. The relative electrical conductance of the solution was recorded as 5.2.

Deduce the nature of the acid. Explain your reasoning.

[2 marks]

QUESTION 7 (7 marks)

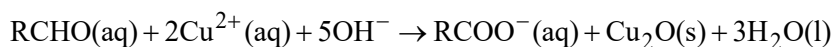
- a) The dissociation constant, K_a , of hypochlorous acid (HOCl) is 3.0×10^{-8} . The conjugate base of HOCl is the hypochlorite ion (OCl^-).

Show that the dissociation constant, K_b , of OCl^- is 3.3×10^{-7} . Assume that the temperature is 25°C .

[2 marks]

QUESTION 8 (7 marks)

Fehling's solution is a deep blue alkaline solution containing a complex of the Cu^{2+} ion. It can be used as a test to detect the presence of glucose in urine, differentiating between the presence of aldehydes and ketones in carbohydrates according to the following equation.



When a reducing sugar is present, the reaction produces a brick-red coloured precipitate of the $\text{Cu}^{+}(\text{I})$ oxide.

The following experiment was set up to test four solutions. Fehling's solution was added to each solution and the solutions were then heated. The observations are recorded in the table.

Solution tested	Observation
glucose	The solution darkened slightly, changing to a brown colour.
sucrose	The solution remained blue.
sucrose and citric acid	The solution changed to a very dark blue with a precipitate starting to form.
starch	The solution remained blue.

- a) i) State why the glucose solution reacted with Fehling's solution. [1 mark]

- ii) State why the sucrose solution did not react with Fehling's solution. [1 mark]

- iii) State why the starch solution did not react with Fehling's solution. [1 mark]

- b) Explain why glucose is sometimes referred to as a reducing sugar. In your response, refer to the reaction between the glucose solution and Fehling's solution. [2 marks]

- c) If the sucrose solution had been hydrolysed before the Fehling's solution was added to it, predict whether a change would have been observed. *[1 mark]*

- d) Infer why a change was observed in the reaction that used the sucrose and citric acid solution. *[1 mark]*

END OF PAPER

